



DATABASE SYSTEMS

(CS-151)

Topic: Database Environment

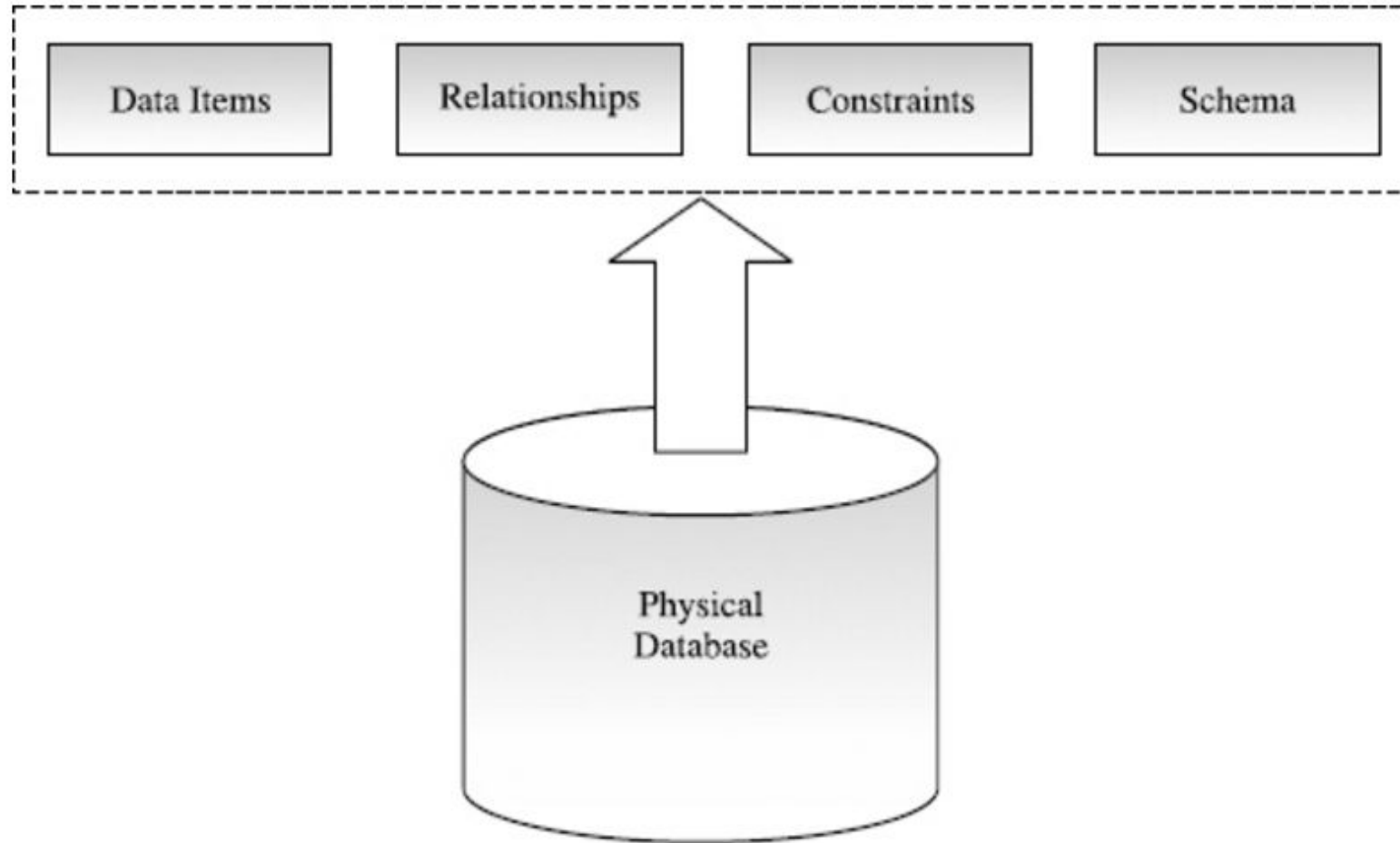
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Lecture Overview

- Database Architecture
- Data Independence
- Different aspects of DBMS

Components of Database



Data Abstraction

Database Systems provide users with an abstract view of data, hiding certain details of how data are stored and maintained

Data Abstraction

- Physical Level

- *Describes how data is actually stored*

- Logical Level

- *Describes what data are stored in the database and what relationships exist among those data*

- View Level

- *Describes only part of the entire database by hiding details of data types.*
 - *Views can also hide information (e.g., salary) for security purposes*

Level of Abstraction

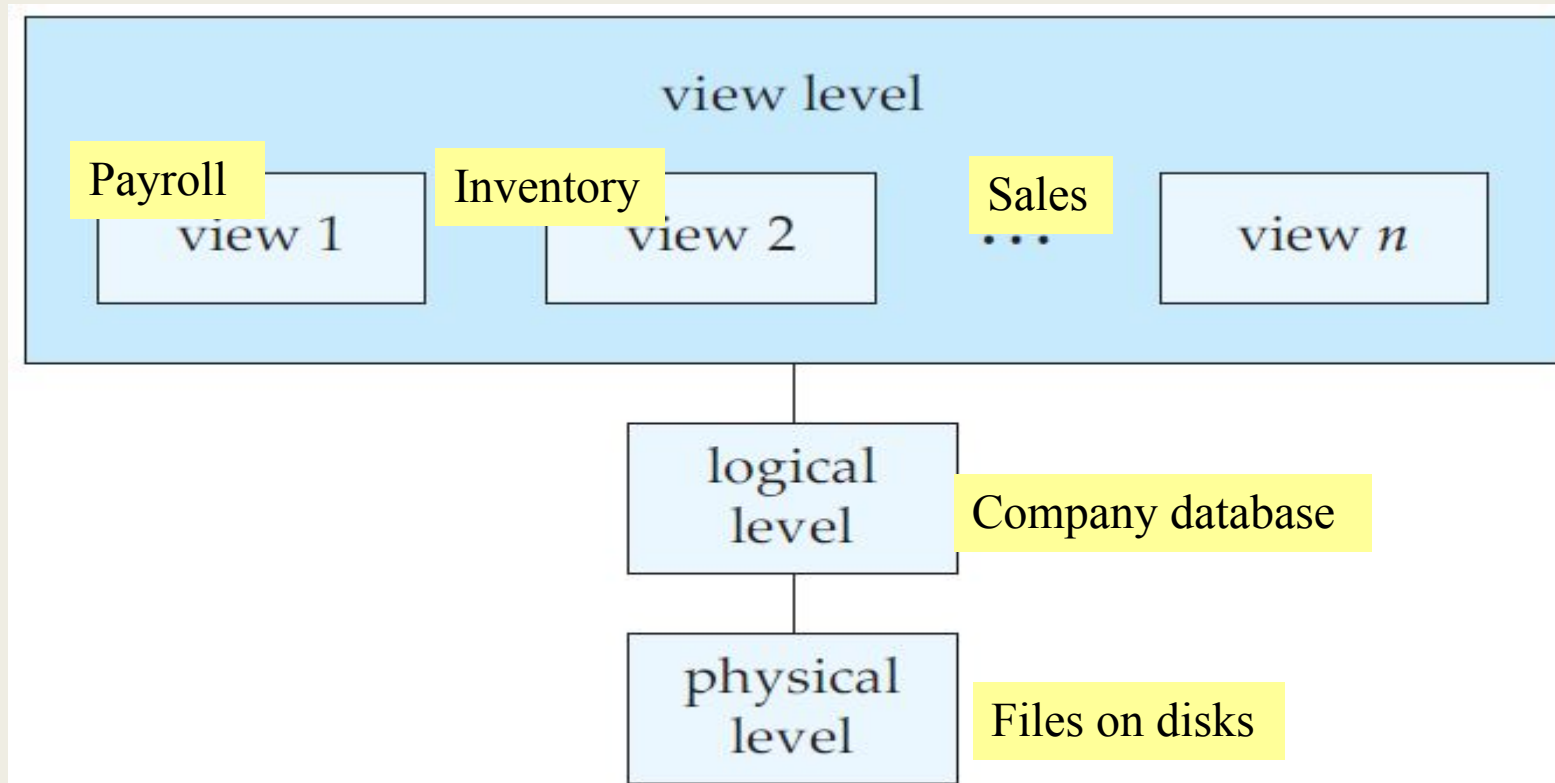


Figure 1.1 The three levels of data abstraction.

Data Item or Field

- A data item is the smallest unit of data that has meaning to its user
- A field consists of a grouping of characters. A data field represents an attribute (a characteristic or quality) of some entity (object, person, place, or event)
- A data item may be used to construct other, more complex structures

Records

- A record represents a collection of attributes that describe a real-world entity.
- A record consists of fields, with each field describing an attribute of the entity.
- Data items are grouped together to form a record.
- Retrieved or updated using programs.

Files

- A file is a collection of related sequence of records
- Files are frequently classified by the application for which they are primarily used (employee file).
- A primary key in a file is the field (or fields) whose value identifies a record among others in a data file.
- Fixed-length records: Every record in the file has exactly the same size
- Variable-length records: different records in the file has different sizes.

Data Dictionary

- A database is a collection of related data that can be queried.
- A data dictionary is an organized list of the structure and attributes of the data stored in a database.
- Data Dictionary are mini database management systems that manage metadata
- The data dictionary is an integral part of the database management systems (DBMSs) and stores metadata, or information about the database, attribute names and definitions for each table in the database.
- Data dictionaries aid the database administrator in the management of a database, user view definitions as well as their use.

Commands

What

Who

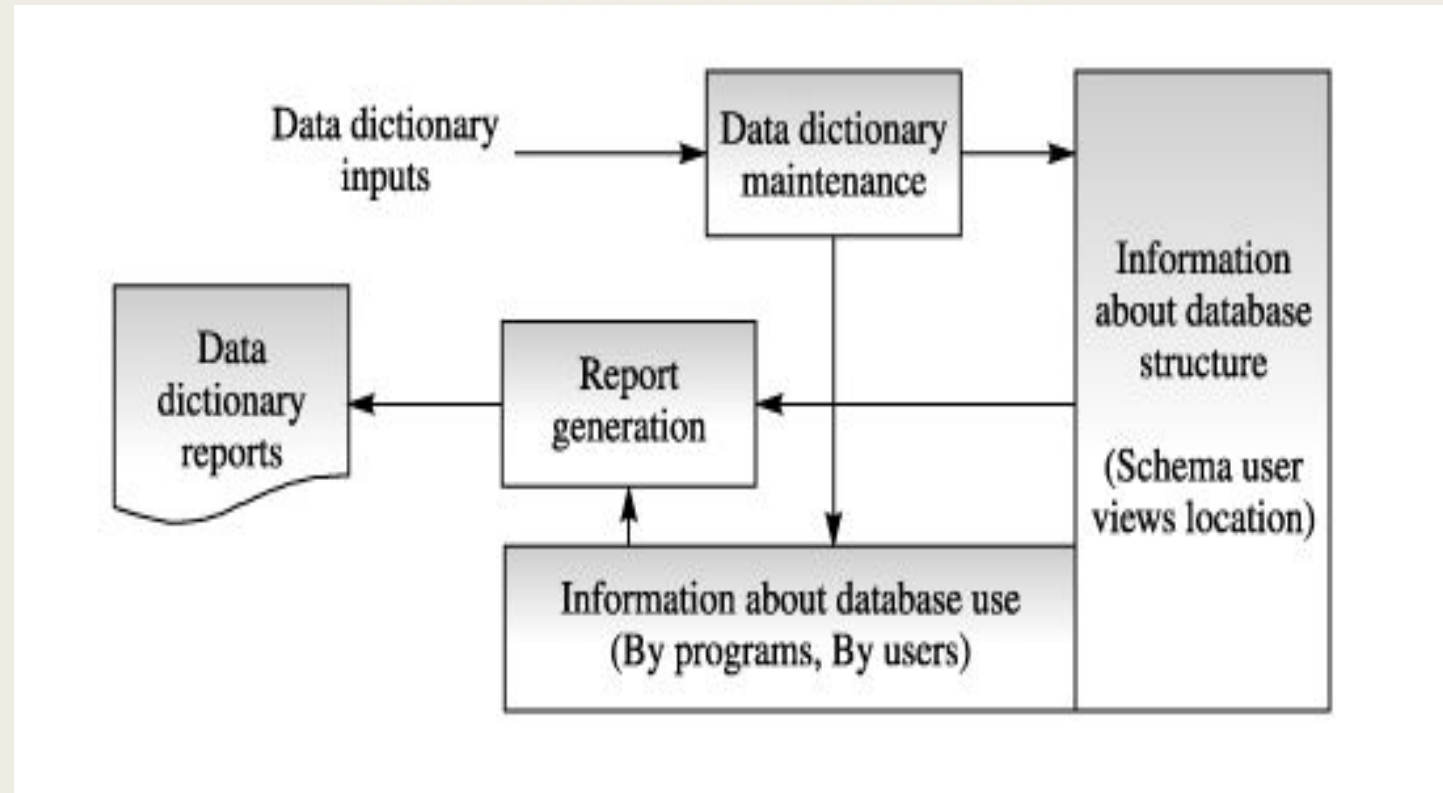
Where

Data Dictionary

- Data Dictionary is made up of two words, data which means the collected information through multiple sources, and dictionary meaning the place where all this information is made available.
- A data dictionary is a crucial part of a relational database as it provides additional information about the relationships between multiple tables in a database. The data dictionary in DBMS helps the user to arrange data in a neat and well-organized way, thus preventing data redundancy.
- Updates when perform (Create, Alter, Drop , Rename)
- Below is a data dictionary describing the table containing employee details.

Attribute Name	Data Type	Max Field Size	Description	isRequired
Employee ID	Integer	10	A unique ID for each Employee	Yes
Name	Text	25	Name of the Employee	Yes
Date of Birth	DateTime	10	Date of Birth of the Employee	Yes
Mobile Number	Integer	10	Contact Number of the Employee	Yes

Structure of Data Dictionary



Data Dictionary

- Data Dictionary contain the following information
 - Descriptions of the schema of the database.
 - Detailed information on physical database design
 - Description of the database users, their responsibilities and their access rights.
 - High-level descriptions of the database transactions and applications and of the relationships of users to transactions.
 - The relationship between database transactions and the data items referenced by them.
 - Usage statistics such as frequencies of queries and transactions and access counts to different portions of the database.

Components of Data Dictionary

- Data Dictionary contain the following components
 - Entities: Any item about which information is stored (record describe an entity, and file describe an entity set)
 - Attributes: property or characteristics of an entity
 - Relationship: Associations that different entities related to each other (For example, a student can take many subjects, while many students take same subject)
 - Keys: Data item for which a computer uses to identify a record in a database system is referred to as key. In other words, key is a single attribute or combination of attributes of an entity set that is used to identify one or more instances of the set.
 - Constraints
 - Access
 -

Data Dictionary

- Passive Data Dictionary
 - Also called non integrated data dictionary
 - Used only for documentation purpose
 - Managed by the user of the system and is modified whenever the structure of the database is changed
 - Modification must be performed by the user manually, it is possible that data dictionary will not be current with the current structure of database

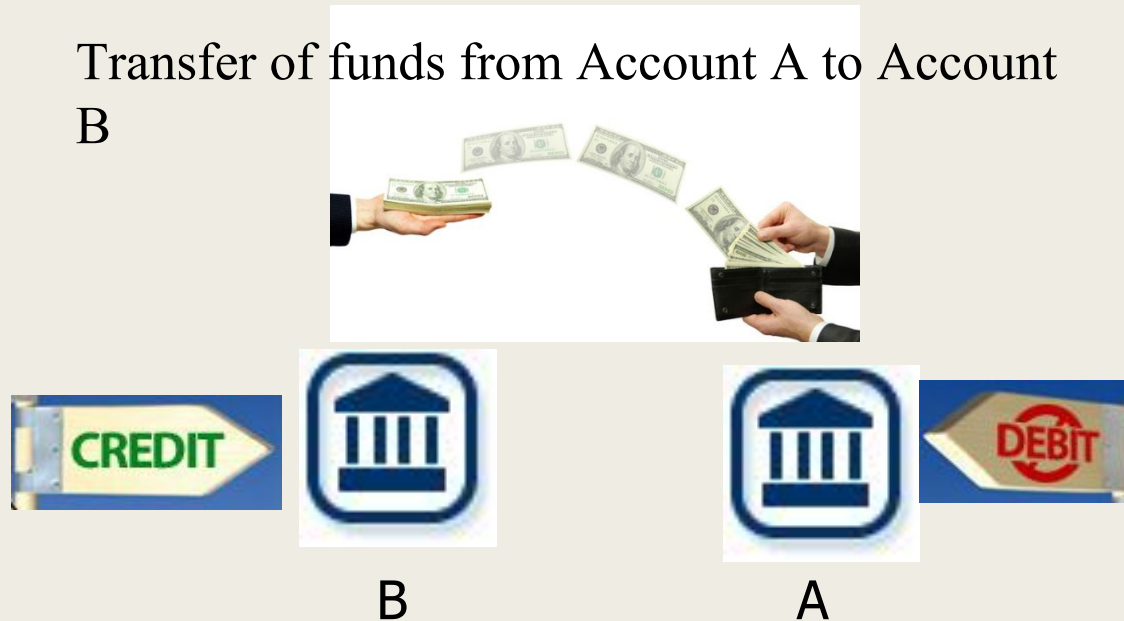
Data Dictionary

- Active Data Dictionary
 - Also called integrated data dictionary
 - Managed automatically by database management software
 - Maintained by the system itself
 - Always consistent with the current structure and definition of the database

Transaction Management

A transaction is a collection of operations that performs a single logical function in a database application

Transfer of funds from Account A to Account B



Transaction Management (ACID)

- Atomicity
 - *All-or-none (Both debit and credit or neither)*
- Consistency
 - *Preserve the consistency of the database ($A+B$ before = $A+B$ after)*
- Isolation
 - *Transaction not to be influenced by changes made by other concurrently executing transactions*
 - *If two users try to modify the same bank account balance simultaneously, isolation ensures that each transaction is executed independently, preventing conflicts.*
- Durability
 - *New values must persist despite system failures (e.g., power failures and operating system crashes)*
 - *Ensures that once a transaction is committed, its changes persist,*
 - *If you transfer money from one bank account to another and the system crashes **after** the transaction is committed, the new balance remains intact once the system is back online.*

Transaction Management

- Each transaction is a unit of ACID(Atomicity, Consistency, Isolation, Durability)
- Plays an important role in shaping many DBMS capabilities, including concurrency control, back up and recovery.

